

Network Tariff Guide for Standard Control Services



1 July 2016 to 30 June 2017



Revision history

Version	Date	Summary of changes
1.0	1 July 2016	Initial 2016–17 Network Tariff Guide.

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1. Introduction

1.1 Overview

Ergon Energy Corporation Limited (Ergon Energy) is a Distribution Network Service Provider (DNSP) to around 730,000 customers in regional Queensland. Our service area covers around 97 per cent of Queensland and has approximately 160,000 kilometres of power lines and one million power poles. Around 70 per cent of the network's power lines are radial and service mostly rural areas with very low levels of customers per line kilometre.

Ergon Energy provides a number of different services. The Australian Energy Regulator (AER) decides how these services are classified and how they are regulated in its Distribution Determination. This is important as it determines how prices are set and how charges are recovered from our customers.

For the 2015 to 2020 period, many of our services are classified as Direct Control Services. These services are subject to direct regulatory oversight by the AER, through price or revenue setting. Direct Control Services are further classified into Standard Control Services and Alternative Control Services.

Standard Control Services are core distribution services associated with the access and supply of electricity to customers. They include network services (e.g. construction, maintenance and repair of the network), some connection services (e.g. small customer connections) and Type 7 metering services. Ergon Energy recovers our costs in providing Standard Control Services through network tariffs billed to retailers.

Alternative Control Services are comprised of:

- *Fee based services* – one-off distribution services that Ergon Energy undertakes at the request of an identifiable customer, retailer or appropriate third party which are levied as a separate charge, in addition to our Standard Control Services. These services are priced on a 'fixed fee' basis as the costs of providing the service (and therefore price) can be assessed in advance of the service being requested.

Examples of fee based services include Type 5 and 6 meter installation and provision (on or after 1 July 2015) during business hours,¹ de-energisations, re-energisations and supply abolishment.

- *Quoted services* – similar to fee based services, but they are 'priced on application' as the nature and scope of these services are variable and the costs (and therefore price) are specific to the individual requestor's needs (e.g. design and construction of connection assets for major customers, real estate development connections and special meter reads etc.).
- *Default Metering Services* – relate to:
 - Type 5 and 6 meter installation and provision (before 1 July 2015)
 - Type 5 and 6 meter installation and provision (on or after 1 July 2015), where the replacement meter is initiated by Ergon Energy as a DNSP
 - Type 5 and 6 metering maintenance, reading and data services.

Ergon Energy recovers our costs of providing Default Metering Services through daily capital and non-capital charges based on the number and type of meters we provide the customer. These charges are billed to retailers.

- *Public Lighting Services* – relate to the provision, construction and maintenance of public lighting assets owned by Ergon Energy, and emerging public lighting technology. Ergon Energy recovers

¹ Where the new or upgraded meter is required as a result of a customer request.

our costs of providing Public Lighting Services through a daily public lighting charge billed to retailers. We also charge a one-off exit fee, when a customer requests the replacement of an existing public light for a light emitting diode (LED) luminaire before the end of its useful operational life.²

We also pass on transmission-related costs and jurisdictional scheme amounts to customers.

1.2 Purpose

This Tariff Guide sets out the network tariffs that apply in 2016–17, as well as the application rules. It is structured as follows:

Section	Description
Section 2	• Outlines what network tariffs are and key pricing concepts • Summarises changes that apply to our network tariffs from 1 July 2016
Appendix 1	• Provides a website link to our 2016–17 Network Tariff Codes and rates (GST exclusive) • Outlines the process for obtaining site specific rates and other information • Sets out explanatory notes on the application of our network tariffs, including tariff conditions • Includes information on Distribution Loss Factors (DLFs) and the Queensland Government's Solar Bonus Scheme (SBS)
Appendix 2	• Shows the tariff classes, network charge categories and Network Tariff Codes applying in 2016–17
Appendix 3	• Summarises the 2016–17 network tariffs, by customer group
Appendix 4	• Outlines the methodology used by Ergon Energy to calculate our Inclining Block Tariff (IBT) network tariffs for Standard Asset Customers (SAC) Small and provides illustrative examples
Appendix 5	• Provides examples of the connection unit charge
Appendix 6	• Provides examples of how the Seasonal Time-of-Use Demand (STOUD) tariffs are calculated for different user groups
Appendix 7	• Provides an example of how the excess reactive power charge is applied
Appendix 8	• Details the kVA methodology used to calculate tariffs for Individually Calculated Customers (ICCs) and Connection Asset Customers (CACs)
Appendix 9	• Lists the Transmission Connection Points (TCPs), their assigned Transmission Node Identifier (TNI) and geographical Transmission Use of System (TUOS) Region

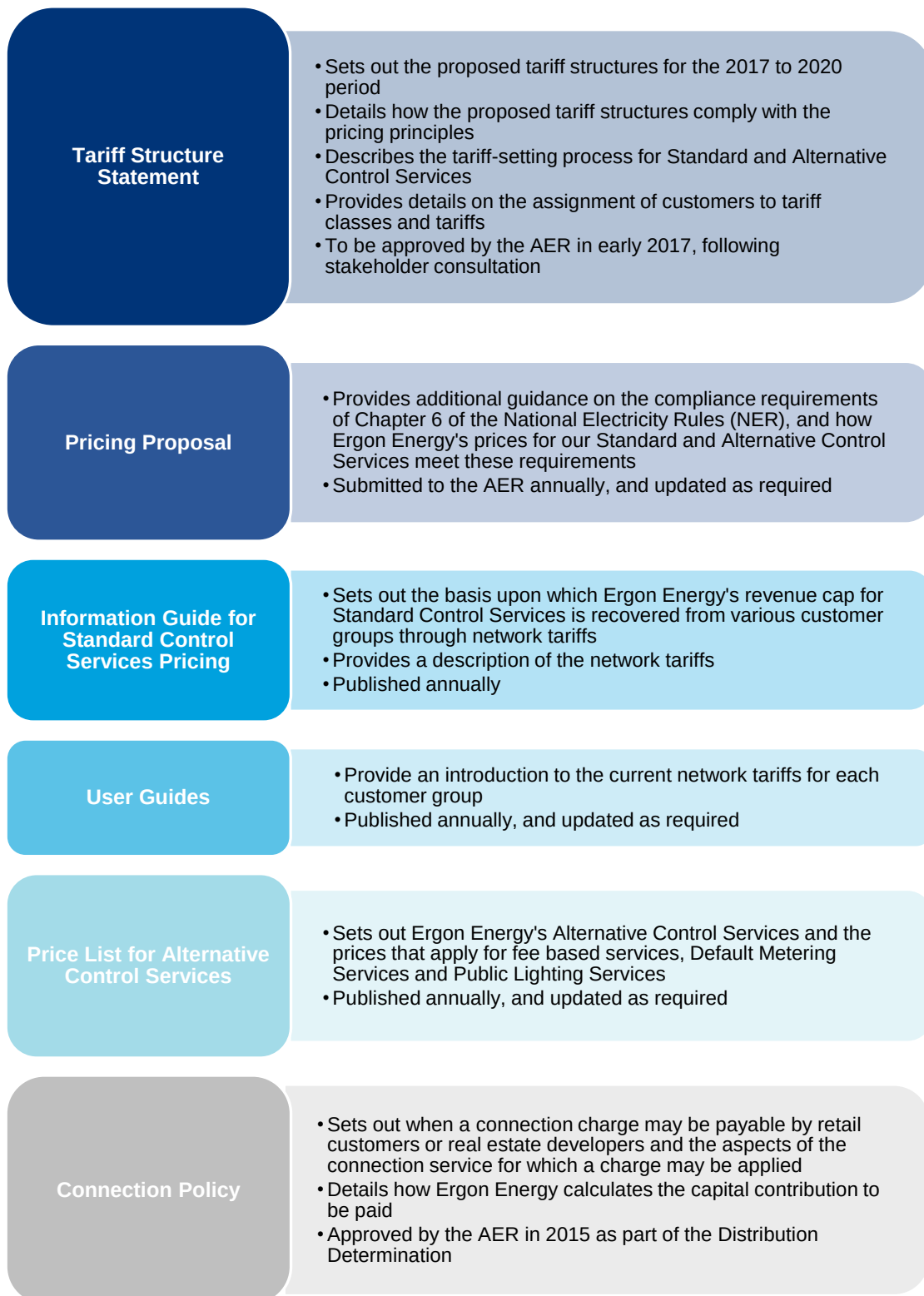
1.3 Supporting network pricing documents

In addition to this Tariff Guide, Ergon Energy has a number of network pricing documents to assist network users, retailers and interested parties understand the development and application of network tariffs, Alternative Control Services prices and connection charges. These documents are outlined in Figure 1.1 below and are available on Ergon Energy's website at:

<https://www.ergon.com.au/network/network-management/network-pricing>.

² Outside of our LED transition program.

Figure 1.1: Supporting network pricing documentation



2. Understanding network tariffs

2.1 Network tariffs

Network tariffs are comprised of:

- Distribution Use of System (DUOS) charges
- TUOS charges
- jurisdictional scheme charges.

2.1.1 DUOS

The DUOS component recovers costs associated with some connection services and/or use of the distribution network for the conveyance of electricity (i.e. Standard Control Services).

Ergon Energy's Standard Control Services are regulated under a revenue cap form of price control. The revenue cap (or 'Total Annual Revenue') for any given year reflects Ergon Energy's smoothed revenue requirement (as determined by the AER), plus other annual revenue adjustments.

The revenue cap is allocated to the three pricing zones and the zonal costs are apportioned to different asset categories within each zone. The costs within the zones are then assigned to the four network user groups and converted into network tariffs that recover the costs.

Further information on the tariff-setting process is available in our *Information Guide for Standard Control Services* (see Section 1.3).

2.1.2 TUOS

Ergon Energy is able to recover transmission-related costs associated with:

- the use of Powerlink's transmission network to deliver high voltage electricity from generators to our network
- Avoided TUOS charges paid to eligible Embedded Generators (EGs)
- payments made to other DNSPs for the supply of distribution services. For Ergon Energy, this includes our connection to Energex Limited's network at Postman's Ridge.

Attachment 14 of the Distribution Determination also allows us to pass through:

- charges levied on Ergon Energy for use of the 220 kV network which supplies the Cloncurry network in our Mount Isa Zone
- entry and exit services charged by Powerlink at three connection points – Stoney Creek, Kings Creek and Oakey Town.

These costs are recovered from customers through TUOS charges.

2.1.3 Jurisdictional scheme

Jurisdictional schemes are certain programs implemented by state governments that place legislative obligations on DNSPs. For Queensland, this includes:

- the SBS, which obligates Ergon Energy to pay a feed-in tariff (FiT) for energy supplied into our distribution network from specific micro-embedded generators
- the energy industry levy. Ergon Energy is obligated under our Distribution Authority to pay a proportion of the Queensland Government's funding commitments for the Australian Energy Market Commission in relation to this levy.

Ergon Energy is able to recover jurisdictional scheme amounts from customers. This is done via separate 'jurisdictional scheme charges'.

2.2 Key concepts in tariff design

There are a number of pricing concepts mentioned in this Tariff Guide. To assist understanding, we have explained these concepts below.

2.2.1 Tariff classes

We have a wide diversity of customers, in terms of their size, location, and usage patterns. We group our customers according to these characteristics. Tariff classes therefore refer to a group of customers with similar characteristics.

Ergon Energy has 18 tariff classes for Standard Control Service customers (see Appendix 2). These tariff classes are differentiated by:

- customer groups
 - ICC
 - CAC
 - SAC
 - SAC Large
 - SAC Small
 - SAC Unmetered
 - EG
- locational zones
 - East Zone
 - West Zone
 - Mount Isa Zone.³

2.2.2 Tariffs, charges and charging parameters

A tariff represents the combination of charges that Ergon Energy applies to a customer (through their retailer) in order to recover network costs. Within each tariff class a number of tariffs can be offered.

Tariffs have three key defining characteristics:

- the charge (can also be called 'charging component', 'tariff component' or 'tariff element')
- the parameters of the charge (specific characteristics that relate to the charge that influence how it is calculated)
- the rate applied to each charge.

Each tariff has at least one charge, but usually has more than one. There are six broad types of charges (e.g. fixed charge) and eight charging parameters (e.g. dollars per day) for our Standard Control Services. Charges and charging parameters are applied differently for each tariff class. Each charge and charging parameter is selected and structured to provide signals to network users about the efficient use of the network.

Other charges and charging parameters apply to the TUOS and jurisdictional scheme components of the network tariff.

³ A zonal map can be found in Ergon Energy's *Information Guide for Standard Control Services Pricing*.

A summary of the charges and charging parameters applying to each network tariff, by customer group, is provided in Appendix 3. This appendix also includes the threshold/minimum chargeable demands for each network tariff, where applicable.

Appendix 1 sets out the rates that apply to each charge.

2.2.3 Network Tariff Codes

Network Tariff Codes are alpha-numeric descriptors used to identify a particular network tariff. In general, each Network Tariff Code has three components:

- a pricing zone code (e.g. “E” represents the East Zone)
- a tariff code (e.g. “BIB” represents the IBT Business tariff)
- a Transmission Region code (e.g. “T1” represents TUOS Region 1).

For example, “EBIBT1” refers to the East Business Inclining Block – TUOS Region 1 network tariff.

A complete listing of our Network Tariff Codes is provided on the ‘List of Network Tariff Codes’ tab in the Appendix 1 spreadsheet.

It should be noted that multiple Network Tariff Codes apply to some SAC network tariffs. This is to enable the billing of the Alternative Control Service ongoing metering charges (i.e. Default Metering Services). Specifically:

- For customers who had a Type 1 to 4 meter installed before 1 July 2015, the Network Tariff Code published in the rates table contains an additional ‘C’. This means that no regulated metering charges will apply.
- For customers who switch to a Type 1 to 4 meter on or after 1 November 2015, the Network Tariff Code published in the rates table contains an additional ‘X’. This means the regulated capital metering charge will apply.

The same network tariff rates listed in Appendix 1 apply to these additional sets of Network Tariff Codes.

2.3 Network tariff changes

Ergon Energy has introduced a number of changes to our network tariff structures from 1 July 2016 for our Standard Control Services. These changes are set out in Table 2.1 below.

Table 2.1: Summary of network tariff changes in 2016–17

Network user group	Tariff changes
CAC	The following changes have been made to the CAC STOUT tariffs: • The peak demand charge is now based on the customer’s monthly maximum kVA demand during the peak period in each summer month. Previously, this charge was based on the greater of the authorised demand and monthly maximum demand during the peak period. • The capacity charge is now based on the greater of the customer’s authorised demand and the actual monthly half hour maximum demand. It will apply for all 12 months of the year. Over the summer months, it excludes demand occurring during the peak demand window of 10.00 am to 8.00 pm on summer weekdays. In 2015–16, this charge was calculated on the greater of a monthly floor and the monthly maximum demand during the non-summer months. We also intended to introduce an excess reactive power charge (excess kVAr charge) in 2016–17 for our CAC network tariffs. Our Pricing Proposal was amended on 30 May 2016 to defer this introduction to 1 July 2017. For network billing purposes,

Network user group	Tariff changes
	the excess kVAr charging component (NDERPC) for CACs will appear as a line item on Ergon Energy's statement of charges to retailers from 1 July 2016 with a \$0.00 rate.
SAC Large	- Continue phasing out the Demand High Voltage tariffs - Progressively increasing the proportion of Long Run Marginal Cost (LRMC) incorporated in the peak demand charge to strengthen the cost reflectivity of the STOUT tariffs.
SAC Small	- In 2016–17, the shoulder and peak energy charges (including times) for the STOUT tariffs (Residential and Business) will be consolidated into one peak energy charge. For network billing purposes, the shoulder component of the STOUT tariffs will be removed and no line item for the NDVCS charging component will appear on bills from 1 July 2016. - The rate for the first 1,000 kWh of annual consumption for the IBT (Residential and Business) has been changed to a positive rate, to progressively align this network tariff with the LRMC pricing principle. - In response to feedback on simplifying the STOUT tariffs (Residential and Business), we have amended the calculation of the peak and off-peak demand charges. The monthly demand charges, for both summer and non-summer, are based on the average demand the customer places on the network in the relevant daily demand window. Previously, the off-peak demand charge was based on the single half-hour maximum demand recorded for the month. Further, we will look at the highest four demand days in the month, determined by the average demand recorded in the relevant daily demand window. The monthly demand rate will then be applied to the average of these top four demand days. In 2015–16, the peak demand charge in the summer months was calculated by identifying the four days in the month with the highest single half-hour peak period demand and averaging the half-hourly demand during the peak periods on those days.
EG	New rules will apply for EGs who are also classified as an ICC (or CAC) for their load connection. We have observed that, when charging for the ICC load, the kVA and kVAr component may contain a contribution from the generator. This has the potential to increase the total kVA and excess kVAr billing quantities. It is not the intent of load side kVA charging for demand and excess kVAr to include this generator impact. Therefore, in 2016–17, for the purposes of network billing for loads, we will set export kVAr to zero in any interval when kW are imported into our distribution network.

For further information on these changes refer to our website.⁴

⁴ www.ergon.com.au/futurenetworktariffs.

Appendix 1: 2016–17 network tariffs

1. Published rates

The 2016–17 Network Tariff Codes and rates have been published in a separate document, *Appendix 1: Ergon Energy 2016–17 SCS Network Tariffs*, which is available with this Network Tariff Guide on our website:

www.ergon.com.au/network/network-management/network-pricing/network-tariffs.

2. Site specific rates

As per clause 6.19.2 of the NER, all information about a service applicant or distribution network user used by a DNSP for the purposes of distribution service pricing is confidential information.

Therefore the publication of site specific network tariffs for ICCs and EGs, and site specific details for CACs, would breach the NER and any connection agreements between Ergon Energy and our customers. It may also adversely affect the markets in which our customers operate.

Information on site-specific Network Tariff Codes, rates, connection units and DLFs are available on request by contacting Network Pricing (netprice@ergon.com.au). Ergon Energy will only release this information in the following circumstances:

- the requestor is the customer, or an authorised person who is able to sign on behalf of the customer (e.g. Company Manager)
- the requestor is the current Financially Responsible Market Participant and the request relates to a period of time for which they are responsible for the site
- the customer's written consent has been provided using the "Authority to Release Information to Third Parties Form" available on Ergon Energy's website (see Section 1.3).

3. Explanatory notes

3.1. SAC Small tariffs

Inclining Block Tariff (IBT) – Default

Under the inclining block structure, there are three consumption blocks – Block 1, Block 2 and Block 3 – with different rates applying to the volume charge for each block. Different block sizes will apply depending on whether the customer is classified as Residential or Business.

A fixed charge per day also applies.

IBT Residential

The IBT Residential tariff is the default primary network tariff applying to all SAC Small residential customers.

The following consumption blocks apply for IBT Residential tariffs:

Block	Daily kWh	Annual equivalent kWh
Block 1	<2.74 kWh	<1,000 kWh p.a.
Block 2	2.74 kWh – 16.43 kWh	1,000 kWh p.a. – 6,000 kWh p.a.
Block 3	>16.43 kWh	>6,000 kWh p.a.

IBT Business

The IBT Business tariff is the default primary network tariff applying to all SAC Small business customers.

The following consumption blocks apply for IBT Business tariffs:

Block	Daily kWh	Annual equivalent kWh
Block 1	<2.74 kWh	<1,000 kWh p.a.
Block 2	2.74 kWh – 54.76 kWh	1,000 kWh p.a. – 20,000 kWh p.a.
Block 3	>54.76 kWh	>20,000 kWh p.a.

IBT network tariff calculation methodology

Ergon Energy's IBT network tariff calculation methodology is set out in Appendix 4.

Seasonal Time-of-Use Energy (STOUE) tariffs – Optional

Under the STOUE structure, there are two time periods – peak and off-peak – with different rates for each time period. Different time periods will apply depending on whether the customer is classified as Residential or Business.

A fixed charge per day also applies.

STOUE – Residential

These tariffs are voluntary for SAC Small residential customers. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This is subject to metering considerations.

The following time periods apply for STOUE – Residential tariffs:

Item	Time period
Peak	3:00 pm to 9:30 pm on summer days
Off-peak	All other times

Note: Summer is defined as the months of December, January and February.

STOUE – Business

These tariffs are voluntary for SAC Small business customers. A customer (or their retailer) must request a tariff change to opt in to these tariffs.

The following time periods apply for STOUE – Business tariffs:

Item	Time period
Peak	10:00 am to 8:00 pm on summer weekdays
Off-peak	All other times

Seasonal Time-of-Use Demand (STOUD) tariffs – Optional

Under the STOUD structure, there are two time periods – peak and off-peak – with different rates for each time period. Different time periods will apply depending on whether the customer is classified as residential or business.

An example calculation of how the STOUD is calculated for SAC Small is provided in Appendix 6.

STOUD – Residential

These tariffs are voluntary for SAC Small residential customers. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This is subject to metering considerations.

Only the demand charge includes seasonal and time-of-day dimensions.

The following time periods apply to the demand and capacity components of STOUD – Residential tariffs:

Item	Time period	Conditions
Peak demand (kW)	3:00 pm to 9:30 pm on all summer days	The monthly demand charges, for both summer and non-summer, are based on the average kW demand the customer places on the network in the daily demand window (the 6.5 hour peak period between 3:00 pm and 9:30 pm). The highest four demand days in the month are determined by comparison of the average demand recorded in these daily demand windows. The monthly demand rate is applied to the average of these top four demand days.
Off-peak demand (kW)	3:00 pm to 9:30 pm on all non-summer days	As above. The off-peak demand quantity is subject to a minimum chargeable demand of 3 kW. The off-peak demand charge does not apply in summer months.

A volume charge (kWh) applies across all energy in summer and non-summer months.⁵

STOUD – Business

These tariffs are voluntary for SAC Small business customers. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This is subject to metering considerations.

Only the demand charge includes seasonal and time-of-day dimensions.

The following time periods apply to the demand and capacity components of STOUD – Business tariffs:

Item	Time period	Conditions
Peak demand (kW)	Peak demand period: 10:00 am to 8:00 pm weekdays in summer months	The monthly demand charges, for both summer and non-summer, are based on the average kW demand the customer places on the network in the daily demand window (the 10 hour peak period on weekdays between 10:00 am and

⁵ It should be noted in 'Appendix 1: Ergon Energy 2016-17 SCS Network Tariffs' the volume charge (kWh) appears as a single charging component (NDVC). For network billing purposes, there are volume peak (NDVCP) and volume off-peak (NDVCOP) components. To give effect to the rates approved by the AER, both volume peak and volume off-peak components apply the same rate.

Item	Time period	Conditions
		8:00 pm). The highest four demand days in the month are determined by comparison of the average demand recorded in these daily demand windows. The monthly demand rate is applied to the average of these top four demand days.
Off-peak demand (kW)	Peak demand period: 10:00 am to 8:00 pm weekdays in non-summer months	As above. The off-peak demand quantity is subject to a minimum chargeable demand of 3 kW. The off-peak demand charge does not apply in summer months.

A volume charge (kWh) applies across all energy in summer and non-summer months.⁶

3.2. SAC Large tariffs

Any time demand tariffs

There are four default network tariffs applicable to SAC Large customers:

- Demand High Voltage (DH)
- Demand Large (DL)
- Demand Medium (DM)
- Demand Small (DS).

The network tariff applicable to a customer will be as nominated by the customer or their retailer. DL, DM and DS are available to all SAC Large customers. These categories are self-selecting, with customers' load characteristics determining the optimum category.

Demand High Voltage is not available to new customers or to existing customers in the West and Mount Isa Zones. DL, DM and DS categories are available for customers connected at HV (22/11 kV Line) or LV (415/240 V).

Application of charges

The actual demand charge for any time demand tariffs will be applied to the kW amount by which a customer's actual monthly maximum demand is greater than the demand threshold applicable to the customer's network tariff.

The threshold demands available are:

Network charge category	Threshold demand
Demand High Voltage	400 kW
Demand Large	400 kW
Demand Medium	120 kW
Demand Small	30 kW

A fixed charge per day and a volume charge (dollars per kWh) apply throughout the year.

Seasonal Time-of-Use Demand (STOUD) tariffs – Optional

These tariffs are voluntary for SAC Large customers. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This is subject to metering considerations.

All of the charging parameters, with the exception of the fixed charge, include seasonal and time-of-day dimensions.

The following time periods apply to the volume (energy) component of the SAC Large STOUD tariffs:

Item	Time period
Off-peak energy	Metered consumption at all times during non-summer months

The following time periods apply to the demand components of SAC Large STOUD tariffs:

Item	Time period	Conditions
Peak demand (kW)	10:00 am to 8:00 pm on summer weekdays	20 kW threshold
Off-peak demand (kW)	All times during non-summer months	40 kW threshold

Application of charges

The peak demand calculation uses the highest kW maximum demand in any single half hour at any time during the peak demand period in each summer month. The demand charge will be applied to the kW amount by which a customer's actual monthly maximum demand is greater than the demand threshold applicable to the peak period. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero.

The off-peak demand calculation uses the highest kW maximum demand in any single half hour at any time during the peak demand period in each non-summer month. The demand charge will be applied to the kW amount by which a customer's actual monthly maximum demand is greater than the demand threshold applicable to the off-peak period. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero.

The off-peak volume calculation is based on a \$/kWh rate applied to metered kWh consumption at all times during non-summer months.⁶ For volume TUOS charge, the metered consumption is multiplied by the DLF and then applied to the TUOS volume rate.

An example calculation of how the STOUD is calculated for SAC Large is provided in Appendix 6.

3.3. CAC tariffs

CAC standardised tariffs – Default

There are four standard default tariffs applicable to CACs:

- 22/11 kV Line
- 22/11 kV Bus
- 33 kV
- 66 kV.

⁶ It should be noted for network billing purposes the STOUD tariff also has a volume peak (NDVCP) charge component. The volume peak (metered consumption at all times during summer months) will appear on bills with a \$0.00 rate.

The network tariff applicable to a customer will be nominated by Ergon Energy, based on the voltage at which a customer is connected to the network. CACs will retain site specific information being their authorised demand (AD) and DLF. A site specific number of connection units will also apply.

The demand-based components are in kVA. Ergon Energy's kVA calculation methodology is set out in Appendix 8.

CAC Seasonal Time-of-Use Demand (STOUD) tariff – Optional

These tariffs are voluntary for CACs. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This is subject to metering considerations.

All of the charging parameters, with the exception of the fixed charge, include seasonal and time-of-day dimensions.

The following time periods apply to the volume (energy) component of CAC STOUD tariffs:

Item	Time period
Off-peak energy	Metered consumption at all times during non-summer months

The following time periods apply to the demand components of CAC STOUD tariffs:

Item	Time period	Conditions
Peak demand (kVA)	10:00 am to 8:00 pm on summer weekdays	
Capacity charge (kVA)	All times during non-summer months and all times during summer months excluding demands occurring during the peak window of 10.00 am to 8.00 pm on summer weekdays.	Applies to the higher of authorised demand or actual monthly maximum demand.

Application of charges

The peak demand calculation uses the maximum kVA demand in any single half hour at any time during the peak demand period in each summer month. The capacity charge calculation uses the maximum of authorised kVA demand or the monthly actual kVA maximum demand during the off-peak window which is all times during non-summer months and all times during summer months excluding demands occurring during the peak window of 10:00am to 8:00pm on summer weekdays.

The connection unit calculation applies the connection unit charge (per day) multiplied by the number of connection units as advised individually to each customer. Refer to Appendix 5 for an illustrative example.

The off-peak volume calculation uses total metered kWh consumption at all times during non-summer months.⁷ For the TUOS volume charge, the metered any time consumption is multiplied by the DLF and then applied to the TUOS volume rate.

An example calculation of how the STOUD is calculated for CACs is provided in Appendix 6.

3.4. ICC site-specific tariffs

ICC network tariffs are site specific with demand based components in kVA. An excess reactive power charge applies to these customers. This charge is applicable when a customer uses more than their permissible kVAr. Permissible kVAr is calculated using the customer's authorised demand and compliant power factor.

⁷ It should be noted for network billing purposes the STOUD tariff also has a volume peak (NDVCP) charge component. The volume peak (metered consumption at all times during summer months) will appear on bills with a \$0.00 rate.

Ergon Energy's excess kVAr calculation methodology is set out in Appendix 7 and kVA calculation methodology is set out in Appendix 8.

3.5. Controlled load tariffs

There are two network charge categories applicable to controlled load:

- **Volume Night Controlled** (EVN, WVN or MVN) – applies where specified permanently connected appliances are controlled by network equipment so that supply will be available for a minimum period of eight hours at the absolute discretion of Ergon Energy, but usually between the hours of 10:00 pm and 7:00 am.
- **Volume Controlled** (EVC, WVC or MVC) – applies where specified permanently connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Ergon Energy.

3.6. Unmetered supplies

There are three network charge categories applicable to unmetered supplies:

- **Unmetered Supply** (EVU, WVU or MVU) – applies to small uniform loads approved to be unmetered by Ergon Energy. Includes provision of energy to traffic lights, watchman lights and other types of unmetered public amenities (e.g. illuminated signs, phone boxes and barbeques etc.). It also applies to public light installations where the customer owns and maintains all assets after the customer's connection point to the network (i.e. where Ergon Energy does not provide Public Lighting Services).
- **Unmetered Supply with ACS – Public Lighting Services – Minor Public Light** (EVUMI, WVUMI or MVUMI) – applies to unmetered 'minor' public lighting where Ergon Energy owns and maintains all assets after the customer's connection point to the network.
- **Unmetered Supply with ACS – Public Lighting Services – Major Public Light** (EVUMA, WVUMA or MVUMA) – applies to unmetered 'major' public lighting where Ergon Energy owns and maintains all assets after the customer's connection point to the network.

The network charge category applicable to a customer will be as nominated by the customer or their retailer.

3.7. General requirements

The following general requirements apply to the selection and amendment of network tariffs:

- It is intended that a Network Tariff Code will apply to each meter data stream.
- Network users with multiple network connections will pay network charges for each connection point. This approach is consistent with the National Metering Identifier (NMI) Procedure issued by the Australian Energy Market Operator.
- If requested, Ergon Energy will provide the customer or the customer's retailer with reasonable assistance in identifying the network tariffs which may be applicable to the customer's installation.
- Demand and/or volume are to be measured at each individual network connection point.
- Where applicable, demand based charges will be calculated on a pro rata daily basis for initial and final accounts.
- Where Ergon Energy's load control equipment exists (including associated metering), this may not be disconnected without Ergon Energy's prior written consent.

- Residential and Business customer classifications align with the Electricity Distribution Network Code (EDNC).
- In the absence of any other customer classification indicators, the IBT Business tariff will be the default primary tariff for SAC Small.
- Where a retailer wants to access the network STOUE or STOU D tariff, and is offering a retail time-of-use (TOU) tariff that differs in the peak, shoulder and off-peak times from the applicable network TOU tariff, then this will be considered a special arrangement. The costs of this special arrangement will be the retailer's responsibility.
- For the purposes of network billing for the IBT, the energy (kWh) assigned to each consumption block during a meter reading period is pro-rated back to a daily equivalent with the tariff applying on a daily basis. The IBT network tariff calculation methodology is set out in Appendix 4.
- Network charges will be billed by Ergon Energy on a monthly basis.
- The TNI identifies the TCP from which a customer is supplied. The NMI Discovery process (Stage 2) will provide retailers with the TNI applicable to a NMI.

3.8. TUOS regional indicators

TUOS charges are charged to Ergon Energy by Powerlink at each TCP or Bulk Supply Point. A TNI is allocated to each of these Bulk Supply Points. TUOS charges for ICCs are directly reflective of the TNI to which they are supplied. For all other classes, charges for each Bulk Supply Point are allocated to one of three geographical TUOS Regions (i.e. T1, T2 or T3).

Appendix 9 lists all the TCPs, their assigned TNI and their geographical TUOS Region.

3.9. Distribution Loss Factors (DLFs)

The standard DLFs that apply in 2016–17 are detailed in the table below.

Network Level	East Zone *		West Zone		Mount Isa Zone	
	Code	DLF	Code	DLF	Code	DLF
Sub-transmission Bus	GESB	1.006	GWSB	1.028	GMSB	1.001
Sub-transmission Line	GESL	1.011	GWSL	1.062	GMSL	1.006
22/11 kV Bus	GEHB	1.015	GWHB	1.068	GMHB	1.008
22/11 kV Line	GEHL	1.028	GWHL	1.100	GMHL	1.036
LV Bus	GELB	1.074	GWLB	1.154	GMLB	1.065
LV Line	GELL	1.095	GWLL	1.185	GMLL	1.069

* The zonal boundaries are identical to those used for network tariffs.

3.10. Public Lighting

Public lighting installations are subject to the same DUOS and TUOS charges applicable to other unmetered supplies.

Alternative Control Service charges also apply. Public Lighting Services is an Alternative Control Service which relates to the provision, construction and maintenance of public lighting assets owned by Ergon Energy, and emerging public lighting technology.

Ergon Energy recovers our costs in providing Public Lighting Services through a daily public lighting charge which we bill to retailers. We also charge a one-off exit fee, when a customer requests the replacement of an existing public light for a LED luminaire before the end of its useful operational life.

3.11. Solar Bonus Scheme (SBS)

The Queensland Government's SBS pays eligible customers a FiT for the surplus electricity generated from solar photovoltaic (PV) systems and exported to the Queensland electricity grid.

Further information is available on the Department of Energy and Water Supply website.⁸

Network Tariff Codes

The Network Tariff Codes that apply to the SBS are detailed in the table below. Details of retailer funded solar FiT arrangements are available on the Queensland Competition Authority's website.⁹

Network Tariff Code	Description
NVG0	Net Volume Generation 0
NVG1	Net Volume Generation 1
NGV2	Net Volume Generation 2
GVG0	Gross Volume Generation 0
NVM0	Net Volume Micro-generation 0

Where:

- NVG0 is applicable to any solar (only) net metering installation that does not meet the SBS requirements
- NVG1 applies to any net metering installation that meets the SBS requirements for the 44 c/kWh FiT rate
- NVG2 applies to any net metering installation that meets the SBS requirements (other than the 44 c/kWh FiT tariff rate)
- GVG0 is applicable to any gross metering installation
- NVM0 applies to any non-solar micro-embedded generating unit with net metering. Customers on this arrangement are not eligible for the SBS.

Recovery of SBS from customers

The costs of the FiT payments made under the SBS will be recovered from network users as a jurisdictional scheme charge as follows:

- a fixed charge (\$/day) for ICCs
- a fixed charge (\$/day) and a volume charge (\$/kWh) for SACs¹⁰ and CACs.

The costs of the energy industry levy are also included in these charges.

Jurisdictional scheme charges apply to all network tariffs, except unmetered supplies and EGs.

⁸ <https://www.dews.qld.gov.au/energy-water-home/electricity/solar-bonus-scheme/>.

⁹ <http://www.qca.org.au/Electricity/Consumer/Solar-Feed-in-Tariffs>.

¹⁰ A fixed charge does not apply to controlled load tariffs.

Appendix 2: Tariff classes, network charge categories and Network Tariff Codes

Tariff classes, network charge categories and Network Tariff Codes 2016–17									
Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
ICC	Individually Calculated Customer - East	ICC - East	ICC				EICCKA1 onwards EICCKB1 onwards	Individually Calculated Customer - East	Individually Calculated Customer - East
	Individually Calculated Customer - West	ICC - West	ICC				WICCKA1 onwards WICCKB1 onwards	Individually Calculated Customer - West	Individually Calculated Customer - West
	Individually Calculated Customer - Mount Isa	ICC - Mount Isa	ICC				MICCKA1 onwards MICCKB1 onwards	Individually Calculated Customer - Mount Isa	Individually Calculated Customer - Mount Isa
CAC	Connection Asset Customer - East	CAC - East	CAC				Site specific	Connection Asset Customer - East	Connection Asset Customers - East
							EC66T1 EC66T2 EC66T3	East Connection Asset Customer 66 kV - TUOS Region 1 East Connection Asset Customer 66 kV TUOS Region 2 East Connection Asset Customer 66 kV - TUOS Region 3	East CAC 66 kV T1 East CAC 66 kV T2 East CAC 66 kV T3
							EC33T1 EC33T2 EC33T3	East Connection Asset Customer 33 kV - TUOS Region 1 East Connection Asset Customer 33 kV - TUOS Region 2 East Connection Asset Customer 33 kV - TUOS Region 3	East CAC 33 kV T1 East CAC 33 kV T2 East CAC 33 kV T3

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							EC22BT1	East Connection Asset Customer 11-22 kV Bus - TUOS Region 1	East CAC 11 22 kV Bus T1
							EC22BT2	East Connection Asset Customer 11-22 kV Bus - TUOS Region 2	East CAC 11 22 kV Bus T2
							EC22BT3	East Connection Asset Customer 11-22 kV Bus - TUOS Region 3	East CAC 11 22 kV Bus T3
							EC22LT1	East Connection Asset Customer 11-22 kV Line - TUOS Region 1	East CAC 11 22 kV Line T1
							EC22LT2	East Connection Asset Customer 11-22 kV Line - TUOS Region 2	East CAC 11 22 kV Line T2
							EC22LT3	East Connection Asset Customer 11-22 kV Line - TUOS Region 3	East CAC 11 22 kV Line T3
							EC66TOUT1	East Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 1	East CAC 33 66 kV ToU Demand T1
							EC66TOUT2	East Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 2	East CAC 33 66 kV ToU Demand T2
							EC66TOUT3	East Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 3	East CAC 33 66 kV ToU Demand T3
							EC22BTOUT1	East Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 1	East CAC 11 22 kV Bus ToU Demand T1
							EC22BTOUT2	East Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 2	East CAC 11 22 kV Bus ToU Demand T2

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							EC22BTOUT3	East Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 3	East CAC 11 22 kV Bus ToU Demand T3
							EC22LTOUT1	East Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 1	East CAC 11 22 kV Line ToU Demand T1
							EC22LTOUT2	East Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 2	East CAC 11 22 kV Line ToU Demand T2
	Connection Asset Customer - West	CAC - West	CAC				EC22LTOUT3	East Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 3	East CAC 11 22 kV Line ToU Demand T3
							WC66T1	West Connection Asset Customer 66 kV - TUOS Region 1	West CAC 66 kV T1
							WC66T2	West Connection Asset Customer 66 kV - TUOS Region 2	West CAC 66 kV T2
							WC66T3	West Connection Asset Customer 66 kV - TUOS Region 3	West CAC 66 kV T3
							WC33T1	West Connection Asset Customer 33 kV - TUOS Region 1	West CAC 33 kV T1
							WC33T2	West Connection Asset Customer 33 kV - TUOS Region 2	West CAC 33 kV T2
							WC33T3	West Connection Asset Customer 33 kV - TUOS Region 3	West CAC 33 kV T3
							WC22BT1	West Connection Asset Customer 11-22 kV Bus - TUOS Region 1	West CAC 11 22 kV Bus T1

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							WC22BT2	West Connection Asset Customer 11-22 kV Bus - TUOS Region 2	West CAC 11 22 kV Bus T2
							WC22BT3	West Connection Asset Customer 11-22 kV Bus - TUOS Region 3	West CAC 11 22 kV Bus T3
							WC22LT1	West Connection Asset Customer 11-22 kV Line - TUOS Region 1	West CAC 11 22 kV Line T1
							WC22LT2	West Connection Asset Customer 11-22 kV Line - TUOS Region 2	West CAC 11 22 kV Line T2
							WC22LT3	West Connection Asset Customer 11-22 kV Line - TUOS Region 3	West CAC 11 22 kV Line T3
							WC66TOUT1	West Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 1	West CAC 33 66 kV ToU Demand T1
							WC66TOUT2	West Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 2	West CAC 33 66 kV ToU Demand T2
							WC66TOUT3	West Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 3	West CAC 33 66 kV ToU Demand T3
							WC22BTOUT1	West Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 1	West CAC 11 22 kV Bus ToU Demand T1
							WC22BTOUT2	West Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 2	West CAC 11 22 kV Bus ToU Demand T2

**Tariff classes, network charge categories and Network Tariff Codes
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Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							WC22BTOUT3	West Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 3	West CAC 11 22 kV Bus ToU Demand T3
							WC22LTOUT1	West Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 1	West CAC 11 22 kV Line ToU Demand T1
							WC22LTOUT2	West Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 2	West CAC 11 22 kV Line ToU Demand T2
	Connection Asset Customer - Mount Isa	CAC - Mount Isa	CAC				WC22LTOUT3	West Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 3	West CAC 11 22 kV Line ToU Demand T3
							MC66T4	Mount Isa Connection Asset Customer 66 kV - TUOS Region 4	Mount Isa CAC 66 kV T4
							MC33T4	Mount Isa Connection Asset Customer 33 kV - TUOS Region 4	Mount Isa CAC 33 kV T4
							MC22BT4	Mount Isa Connection Asset Customer 11-22 kV Bus - TUOS Region 4	Mount Isa CAC 11 22 kV Bus T4
							MC22LT4	Mount Isa Connection Asset Customer 11-22 kV Line - TUOS Region 4	Mount Isa CAC 11 22 kV Line T4
							MC66TOUT4	Mount Isa Connection Asset Customer 33-66 kV Time-of-Use Demand - TUOS Region 4	Mount Isa CAC 33 66 kV ToU Demand T4
							MC22BTOUT4	Mount Isa Connection Asset Customer 11-22 kV Bus Time-of-Use Demand - TUOS Region 4	Mount Isa CAC 11 22 kV Bus ToU Demand T4

**Tariff classes, network charge categories and Network Tariff Codes
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Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							MC22LTOUT4	Mount Isa Connection Asset Customer 11-22 kV Line Time-of-Use Demand - TUOS Region 4	Mount Isa CAC 11 22 kV Line ToU Demand T4
EG	Embedded Generation - East	EG - East	EG				EEGA1 onwards EEGB1 onwards	Embedded Generation - East	Embedded Generation - East
	Embedded Generation - West	EG - West	EG				WEGA1 onwards WEGB1 onwards	Embedded Generation - West	Embedded Generation - West
	Embedded Generation - Mount Isa	EG - Mount Isa	EG				MEGA1 onwards MEGB1 onwards	Embedded Generation - Mount Isa	Embedded Generation - Mount Isa
SAC	Standard Asset Customer - Large (>100 MWh p.a.) - East	SAC Large - East	Demand High Voltage	400 kW			EDHTT1	East Demand HV (400 kW) SAC - TUOS Region 1	East Demand HV 400 kW T1
				400 kW			EDHTT2	East Demand HV (400 kW) SAC - TUOS Region 2	East Demand HV 400 kW T2
				400 kW			EDHTT3	East Demand HV (400 kW) SAC - TUOS Region 3	East Demand HV 400 kW T3
			Demand Large	400 kW			EDLTT1	East Demand Large (400 kW) SAC - TUOS Region 1	East Demand LGE 400 kW Threshold T1
				400 kW			EDLTT2	East Demand Large (400 kW) SAC - TUOS Region 2	East Demand LGE 400 kW Threshold T2
				400 kW			EDLTT3	East Demand Large (400 kW) SAC - TUOS Region 3	East Demand LGE 400 kW Threshold T3
			Demand Medium	120 kW			EDMTT1	East Demand Medium (120 kW) SAC - TUOS Region 1	East Demand MED 120 kW Threshold T1
				120 kW			EDMTT2	East Demand Medium (120 kW) SAC - TUOS Region 2	East Demand MED 120 kW Threshold T2

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
				120 kW			EDMTT3	East Demand Medium (120 kW) SAC - TUOS Region 3	East Demand MED 120 kW Threshold T3
			Demand Small	30 kW			EDSTT1	East Demand Small (30 kW) SAC - TUOS Region 1	East Demand SML 30 kW Threshold T1
				30 kW			EDSTT2	East Demand Small (30 kW) SAC - TUOS Region 2	East Demand SML 30 kW Threshold T2
				30 kW			EDSTT3	East Demand Small (30 kW) SAC - TUOS Region 3	East Demand SML 30 kW Threshold T3
			Seasonal TOU Demand		20 kW	40 kW	ESTOUDCT1	East Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 1	East Seasonal ToU Demand Comms Meters T1
					20 kW	40 kW	ESTOUDCT2	East Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 2	East Seasonal ToU Demand Comms Meters T2
					20 kW	40 kW	ESTOUDCT3	East Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 3	East Seasonal ToU Demand Comms Meters T3
	Standard Asset Customer - Small (<100 MWh p.a.) - East	SAC Small - East	IBT Business				EBIBT1	East Business Inclining Block - TUOS Region 1	East Business Inclining Block T1
							EBIBT2	East Business Inclining Block - TUOS Region 2	East Business Inclining Block T2
							EBIBT3	East Business Inclining Block - TUOS Region 3	East Business Inclining Block T3
			IBT Residential				ERIBT1	East Residential Inclining Block - TUOS Region 1	East Residential Inclining Block T1
							ERIBT2	East Residential Inclining Block - TUOS Region 2	East Residential Inclining Block T2
							ERIBT3	East Residential Inclining Block - TUOS Region 3	East Residential Inclining Block T3
			Seasonal TOU Energy Business				EBTOUT1	East Business Time-of-Use - TUOS Region 1	East Business ToU Energy T1
							EBTOUT2	East Business Time-of-Use - TUOS Region 2	East Business ToU Energy T2

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
							EBTOUT3	East Business Time-of-Use - TUOS Region 3	East Business ToU Energy T3
			Seasonal TOU Energy Residential				ERTOUT1	East Residential Time-of-Use - TUOS Region 1	East Residential ToU Energy T1
							ERTOUT2	East Residential Time-of-Use - TUOS Region 2	East Residential ToU Energy T2
							ERTOUT3	East Residential Time-of-Use - TUOS Region 3	East Residential ToU Energy T3
			Seasonal TOU Demand Business			3 kW	EBTOUDCT1	East Business Time-of-Use Demand Comms Metering - TUOS Region 1	East Business ToU Demand Comms Meters T1
						3 kW	EBTOUDCT2	East Business Time-of-Use Demand Comms Metering - TUOS Region 2	East Business ToU Demand Comms Meters T2
						3 kW	EBTOUDCT3	East Business Time-of-Use Demand Comms Metering - TUOS Region 3	East Business ToU Demand Comms Meters T3
			Seasonal TOU Demand Residential			3 kW	ERTOUDCT1	East Residential Time-of-Use Demand Comms Metering - TUOS Region 1	East Residential ToU Demand Comms Meters T1
						3 kW	ERTOUDCT2	East Residential Time-of-Use Demand Comms Metering - TUOS Region 2	East Residential ToU Demand Comms Meters T2
						3 kW	ERTOUDCT3	East Residential Time-of-Use Demand Comms Metering - TUOS Region 3	East Residential ToU Demand Comms Meters T3
			Volume Controlled				EVCT1	East Volume Controlled (T33) - TUOS Region 1	East Volume Controlled (T33) T1
							EVCT2	East Volume Controlled (T33) - TUOS Region 2	East Volume Controlled (T33) T2
							EVCT3	East Volume Controlled (T33) - TUOS Region 3	East Volume Controlled (T33) T3

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
			Volume Night Controlled				EVNT1	East Volume Night Controlled (T31) - TUOS Region 1	East Volume Night Controlled (T31) T1
							EVNT2	East Volume Night Controlled (T31) - TUOS Region 2	East Volume Night Controlled (T31) T2
							EVNT3	East Volume Night Controlled (T31) - TUOS Region 3	East Volume Night Controlled (T31) T3
	Standard Asset Customer - Unmetered - East	SAC Unmetered - East	Volume Unmetered				EVUT1	East Volume Unmetered - TUOS Region 1	East Volume Unmetered T1
							EVUT2	East Volume Unmetered - TUOS Region 2	East Volume Unmetered T2
							EVUT3	East Volume Unmetered - TUOS Region 3	East Volume Unmetered T3
			Volume Unmetered (Minor public light)				EVUMIT1	East Volume Unmetered - Minor Public Light - TUOS Region 1	East Volume Unmetered Minor SL T1
							EVUMIT2	East Volume Unmetered - Minor Public Light - TUOS Region 2	East Volume Unmetered Minor SL T2
							EVUMIT3	East Volume Unmetered - Minor Public Light - TUOS Region 3	East Volume Unmetered Minor SL T3
			Volume Unmetered (Major public light)				EVUMAT1	East Volume Unmetered - TUOS Region 1 - Major Public Light	East Volume Unmetered Major SL T1
							EVUMAT2	East Volume Unmetered - Major Public Light - TUOS Region 2	East Volume Unmetered Major SL T2
							EVUMAT3	East Volume Unmetered - Major Public Light - TUOS Region 3	East Volume Unmetered Major SL T3
	Standard Asset Customer - Large	SAC Large - West	Demand Large	400 kW			WDLTT1	West Demand Large (400 kW) SAC - TUOS Region 1	West Demand LGE 400 kW Threshold T1

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
	(>100 MWh p.a.) - West			400 kW			WDLTT2	West Demand Large (400 kW) SAC - TUOS Region 2	West Demand LGE 400 kW Threshold T2
				400 kW			WDLTT3	West Demand Large (400 kW) SAC - TUOS Region 3	West Demand LGE 400 kW Threshold T3
			Demand Medium	120 kW			WDMTT1	West Demand Medium (120 kW) SAC - TUOS Region 1	West Demand MED 120 kW Threshold T1
				120 kW			WDMTT2	West Demand Medium (120 kW) SAC - TUOS Region 2	West Demand MED 120 kW Threshold T2
				120 kW			WDMTT3	West Demand Medium (120 kW) SAC - TUOS Region 3	West Demand MED 120 kW Threshold T3
			Demand Small	30 kW			WDSTT1	West Demand Small (30 kW) SAC - TUOS Region 1	West Demand SML 30 kW Threshold T1
				30 kW			WDSTT2	West Demand Small (30 kW) SAC - TUOS Region 2	West Demand SML 30 kW Threshold T2
				30 kW			WDSTT3	West Demand Small (30 kW) SAC - TUOS Region 3	West Demand SML 30 kW Threshold T3
			Seasonal TOU Demand		20 kW	40 kW	WSTOUDCT1	West Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 1	West Seasonal ToU Demand Comms Meters T1
					20 kW	40 kW	WSTOUDCT2	West Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 2	West Seasonal ToU Demand Comms Meters T2
					20 kW	40 kW	WSTOUDCT3	West Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 3	West Seasonal ToU Demand Comms Meters T3
	Standard Asset Customer - Small (<100 MWh p.a.) - West	SAC Small - West	IBT Business				WBIBT1	West Business Inclining Block - TUOS Region 1	West Business Inclining Block T1
							WBIBT2	West Business Inclining Block - TUOS Region 2	West Business Inclining Block T2
							WBIBT3	West Business Inclining Block - TUOS Region 3	West Business Inclining Block T3

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
			IBT Residential				WRIBT1	West Residential Inclining Block - TUOS Region 1	West Residential Inclining Block T1
							WRIBT2	West Residential Inclining Block - TUOS Region 2	West Residential Inclining Block T2
							WRIBT3	West Residential Inclining Block - TUOS Region 3	West Residential Inclining Block T3
			Seasonal TOU Energy Business				WBTOU1	West Business Time-of-Use Energy - TUOS Region 1	West Business ToU Energy T1
							WBTOU2	West Business Time-of-Use Energy - TUOS Region 2	West Business ToU Energy T2
							WBTOU3	West Business Time-of-Use Energy - TUOS Region 3	West Business ToU Energy T3
			Seasonal TOU Energy Residential				WRTOUT1	West Residential Time-of-Use Energy - TUOS Region 1	West Residential ToU Energy T1
							WRTOUT2	West Residential Time-of-Use Energy - TUOS Region 2	West Residential ToU Energy T2
							WRTOUT3	West Residential Time-of-Use Energy - TUOS Region 3	West Residential ToU Energy T3
			Seasonal TOU Demand Business			3 kW	WBTOUTDCT1	West Business Time-of-Use Demand Comms Metering - TUOS Region 1	West Business ToU Demand Comms Meters T1
						3 kW	WBTOUTDCT2	West Business Time-of-Use Demand Comms Metering - TUOS Region 2	West Business ToU Demand Comms Meters T2
						3 kW	WBTOUTDCT3	West Business Time-of-Use Demand Comms Metering - TUOS Region 3	West Business ToU Demand Comms Meters T3

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
			Seasonal TOU Demand Residential			3 kW	WRTOUDCT1	West Residential Time-of-Use Demand Comms Metering - TUOS Region 1	West Residential ToU Demand Comms Meters T1
						3 kW	WRTOUDCT2	West Residential Time-of-Use Demand Comms Metering - TUOS Region 2	West Residential ToU Demand Comms Meters T2
						3 kW	WRTOUDCT3	West Residential Time-of-Use Demand Comms Metering - TUOS Region 3	West Residential ToU Demand Comms Meters T3
			Volume Controlled				WVCT1	West Volume Controlled (T33) - TUOS Region 1	West Volume Controlled (T33) T1
							WVCT2	West Volume Controlled (T33) - TUOS Region 2	West Volume Controlled (T33) T2
							WVCT3	West Volume Controlled (T33) - TUOS Region 3	West Volume Controlled (T33) T3
			Volume Night Controlled				WVNT1	West Volume Night Controlled (T31) - TUOS Region 1	West Volume Night Controlled (T31) T1
							WVNT2	West Volume Night Controlled (T31) - TUOS Region 2	West Volume Night Controlled (T31) T2
							WVNT3	West Volume Night Controlled (T31) - TUOS Region 3	West Volume Night Controlled (T31) T3
	Standard Asset Customer - Unmetered – West	SAC Unmetered - West	Volume Unmetered				WVUT1	West Volume Unmetered - TUOS Region 1	West Volume Unmetered T1
							WVUT2	West Volume Unmetered - TUOS Region 2	West Volume Unmetered T2
							WVUT3	West Volume Unmetered - TUOS Region 3	West Volume Unmetered T3
			Volume Unmetered (Minor public)				WVUMIT1	West Volume Unmetered - Minor Public Light - TUOS Region 1	West Volume Unmetered Minor SL T1

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
			light)				WVUMIT2	West Volume Unmetered - Minor Public Light - TUOS Region 2	West Volume Unmetered Minor SL T2
							WVUMIT3	West Volume Unmetered - Minor Public Light - TUOS Region 3	West Volume Unmetered Minor SL T3
			Volume Unmetered (Major public light)				WVUMAT1	West Volume Unmetered - Major Public Light - TUOS Region 1	West Volume Unmetered Major SL T1
							WVUMAT2	West Volume Unmetered - Major Public Light - TUOS Region 2	West Volume Unmetered Major SL T2
							WVUMAT3	West Volume Unmetered - Major Public Light - TUOS Region 3	West Volume Unmetered Major SL T3
	Standard Asset Customer - Large (>100 MWh p.a.) - Mount Isa	SAC Large - Mount Isa	Demand Large	400 kW			MDLTT4	Mount Isa Demand Large (400 kW) SAC - TUOS Region 4	Mount Isa Demand LGE 400 kW Threshold T4
			Demand Medium	120 kW			MDMTT4	Mount Isa Demand Medium (120 kW) SAC - TUOS Region 4	Mount Isa Demand MED 120 kW Threshold T4
			Demand Small	30 kW			MDSTT4	Mount Isa Demand Small (30 kW) SAC - TUOS Region 4	Mount Isa Demand SML 30 kW Threshold T4
			Seasonal TOU Demand		20 kW	40 kW	MSTOUDCT4	Mount Isa Seasonal Time-of-Use Demand SAC Comms Metering - TUOS Region 4	Mount Isa Seasonal ToU Demand Comms Meters T4
	Standard Asset Customer - Small (<100 MWh p.a.) - Mount Isa	SAC Small - Mount Isa	IBT Business				MBIBT4	Mount Isa Business Inclining Block - TUOS Region 4	Mount Isa Business Inclining Block T4
			IBT Residential				MRIBT4	Mount Isa Residential Inclining Block - TUOS Region 4	Mount Isa Residential Inclining Block T4
			Seasonal TOU Energy Business				MBTOUT4	Mount Isa Business Time-of-Use Energy - TUOS Region 4	Mount Isa Business ToU Energy T4

**Tariff classes, network charge categories and Network Tariff Codes
2016–17**

Network user group	Tariff class	Abbreviated Tariff class	Network charge category	Threshold/minimum chargeable demand			Network Tariff Code	Description	Abbreviated Description
				Any time	Peak	Off-peak			
			Seasonal TOU Energy Residential				MRTOUT4	Mount Isa Residential Time-of-Use Energy - TUOS Region 4	Mount Isa Residential ToU Energy T4
			Seasonal TOU Demand Business			3 kW	MBTOUDCT4	Mount Isa Business Time-of-Use Demand Comms Metering - TUOS Region 4	Mount Isa Business ToU Demand Comms Meters T4
			Seasonal TOU Demand Residential			3 kW	MRTOUDCT4	Mount Isa Residential Time-of-Use Demand - TUOS Region 4	Mount Isa Residential ToU Demand T4
			Volume Controlled				MVCT4	Mount Isa Volume Controlled (T33) - TUOS Region 4	Mount Isa Volume Controlled (T33) T4
			Volume Night Controlled				MVNT4	Mount Isa Volume Night Controlled (T31) Comms Metering- TUOS Region 4	Mount Isa Volume Night Controlled (T31) T4
	Standard Asset Customer - Unmetered - Mount Isa	SAC Unmetered - Mount Isa	Volume Unmetered				MVUT4	Mount Isa Volume Unmetered - TUOS Region 4	Mount Isa Volume Unmetered T4
			Volume Unmetered (Minor public light)				MVUMIT4	Mount Isa Volume Unmetered - Minor Public Light - TUOS Region 4	Mount Isa Volume Unmetered Minor SL T4
			Volume Unmetered (Major public light)				MVUMAT4	Mount Isa Volume Unmetered - Major Public Light - TUOS Region 4	Mount Isa Volume Unmetered Major SL T4

Appendix 3: Summary of network tariffs, by network user group

Network user group	Network tariff	Description / tariff conditions	DUOS											TUOS							Jurisdictional Scheme	
			Charging parameters											Charging parameters							Charging parameters	
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge			Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge			Volume charge	Jurisdictional scheme charge	
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVAr/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh
ICC	ICC	Applies to customers classified as an ICC.	✓		✓		✓	✓		✓			✓	✓	✓				✓	✓		
CAC	CAC – 66 kV	Applies to customers connected to either a 66 kV substation or a 66 kV line who are classified as a CAC.	✓	✓	✓			✓		✓			✓	✓					✓	✓	✓	
	CAC – 33 kV	Applies to customers connected to either a 33 kV substation or a 33 kV line who are classified as a CAC.	✓	✓	✓			✓		✓			✓	✓					✓	✓	✓	
	CAC – 22 kV Bus	Applies to customers connected to either a 22 kV or 11 kV substation who are classified as a CAC.	✓	✓	✓			✓		✓			✓	✓					✓	✓	✓	
	CAC – 22 kV Line	Applies to customers connected to either a 22 kV or 11 kV line who are classified as a CAC.	✓	✓	✓			✓		✓			✓	✓					✓	✓	✓	

Network user group	Network tariff	Description / tariff conditions	DUOS										TUOS							Jurisdictional Scheme		
			Charging parameters										Charging parameters							Charging parameters		
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge			Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge			Volume charge	Jurisdictional scheme charge	
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVAr/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh
	CAC - STOUD	Applies to customers classified as a CAC.	✓	✓		✓				✓			✓	✓	✓					✓	✓	✓
EG	EG	Applies to customers classified as an EG. Tariff applies to generation side only. Additional tariffs apply to the load side of an EG.	✓																			
SAC Large	Demand High Voltage	Not available to new customers or existing customers in the West and Mount Isa Zones.																				
		Primary tariff available to high voltage metered customers who are not otherwise classified as an ICC, CAC or EG.	✓						✓		✓			✓				✓		✓	✓	✓
		Threshold demand – 400 kW																				
	Demand Large	Primary tariff available for SAC customers consuming more than 100 MWh p.a.	✓								✓		✓					✓		✓	✓	✓
		Threshold demand – 400 kW																				

Network user group	Network tariff	Description / tariff conditions	DUOS										TUOS							Jurisdictional Scheme		
			Charging parameters										Charging parameters							Charging parameters		
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge			Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge			Volume charge	Jurisdictional scheme charge	
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh	
	Demand Medium	Primary tariff available for SAC customers consuming more than 100 MWh p.a.	✓					✓		✓			✓				✓		✓	✓	✓	
		Threshold demand – 120 kW																✓	✓			
	Demand Small	Primary tariff available for SAC customers consuming more than 100 MWh p.a.	✓					✓		✓			✓				✓		✓	✓	✓	
		Threshold demand – 30 kW																✓	✓			
	Seasonal TOU Demand	Optional STOUT tariff available to SAC customers consuming more than 100 MWh p.a. regardless of their supply voltage (high or low)	✓						✓			✓	✓				✓	✓	✓	✓	✓	
		Threshold demand – 20 kW peak; 40 kW off-peak.																				
SAC Small	IBT Residential	Default primary tariff that applies to residential customers with consumption less than 100 MWh p.a.	✓								✓		✓					✓	✓	✓		

Network user group	Network tariff	Description / tariff conditions	DUOS										TUOS							Jurisdictional Scheme		
			Charging parameters										Charging parameters							Charging parameters		
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge		Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge			Volume charge	Jurisdictional scheme charge		
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVAr/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh
		Three consumption blocks apply to the DUOS volume charge – Block 1, Block 2 and Block 3.																				
	IBT Business	Default primary tariff that applies to business customers with consumption less than 100 MWh p.a.	✓								✓		✓						✓	✓	✓	
		Three consumption blocks apply to the DUOS volume charge – Block 1, Block 2 and Block 3.																				
	Seasonal TOU Energy Residential	Optional primary tariff that applies to residential customers with consumption less than 100 MWh p.a.	✓										✓	✓						✓	✓	✓
		Two time periods apply to the DUOS volume charge – peak and off-peak.																				

Network user group	Network tariff	Description / tariff conditions	DUOS										TUOS						Jurisdictional Scheme			
			Charging parameters										Charging parameters						Charging parameters			
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge			Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge			Volume charge	Jurisdictional scheme charge	
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh	
	Seasonal TOU Energy Business	Optional primary tariff that applies to business customers with consumption less than 100 MWh p.a.	✓									✓	✓						✓	✓	✓	
		Two time periods apply to the DUOS volume charge – peak and off-peak.																				
	Seasonal TOU Demand Residential	Voluntary STOU tariff that applies to residential customers with consumption less than 100 MWh p.a.	✓ ¹¹						✓	✓			✓						✓	✓	✓	
		Daily demand windows apply.																				
	Seasonal TOU Demand Business	Voluntary STOU tariff that applies to business customers with consumption less than 100 MWh p.a.	✓ ¹²							✓	✓			✓						✓	✓	✓
		Daily demand windows apply.																				

¹¹ A \$0.00 rate applies.

¹² A \$0.00 rate applies.

Network user group	Network tariff	Description / tariff conditions	DUOS									TUOS						Jurisdictional Scheme			
			Charging parameters									Charging parameters						Charging parameters			
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge		Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge		Volume charge	Jurisdictional scheme charge		
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh
	Volume Controlled	Secondary tariff available for customers consuming less than 100 MWh p.a. who have loads that are eligible to have supply controlled by Ergon Energy (e.g. hot water, air-conditioning, and pool pumps etc.).	✓							✓									✓		✓
		Supply is generally available for a minimum period of 18 hours each day, at the absolute discretion of Ergon Energy.																			
	Volume Night Controlled	Secondary tariff available for customers consuming less than 100 MWh p.a. who have loads that are eligible to have supply controlled by Ergon Energy.	✓							✓									✓		✓

Network user group	Network tariff	Description / tariff conditions	DUOS									TUOS						Jurisdictional Scheme				
			Charging parameters									Charging parameters						Charging parameters				
			Fixed charge	Connection unit Charge	Capacity charge		Excess reactive power charge	Actual demand charge		Volume charge			Fixed charge	Capacity charge	Common service and general charge	Actual demand charge		Volume charge	Jurisdictional scheme charge			
			\$/day	\$/connection unit/day	\$/kVA of AD/mth	Peak, Off-peak \$/kVA of AD/mth	\$/kVAr/mth	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	Any time \$/kWh/mth	Block 1, Block 2, Block 3 \$/kWh/mth	Peak, Off-peak \$/kWh	\$/day	\$/kVA of AD/mth	\$/day	\$/kVA/mth	\$/kW/mth	Peak, Off-peak \$/kW/mth	\$/kWh	\$/day	\$/kWh
		Supply is generally available for a minimum period of 8 hours each day, at the absolute discretion of Ergon Energy, but usually for a limited number of hours each night in off-peak periods.																				
	Unmetered Supply	Applies to small uniform loads approved to be unmetered by Ergon Energy. This includes the provision of energy to public lights, traffic lights, watchman lights and other types of unmetered public amenities (e.g. illuminated signs, phone boxes and barbeques etc.).	✓ ¹³								✓									✓		

¹³ \$/day/device.

Appendix 4: Ergon Energy's IBT network tariff calculation methodology

The IBT is structured with a fixed rate per day and three energy consumption blocks, each with a different volume (energy) rate applicable.

Block and fixed rates are different between the East, West and Mount Isa Zones and between residential and business customers. Block sizes are also different between residential and business customers.

The IBT is denominated and applied on a daily basis. However, it may be described in the context of an annual basis for network tariff consultation and presentation purposes. For example, the IBT Residential consumption blocks are:

Block	Daily kWh	Annual equivalent kWh
Block 1	<2.74 kWh	<1,000 kWh p.a.
Block 2	2.74 kWh – 16.43 kWh	1,000 kWh p.a. – 6,000 kWh p.a.
Block 3	>16.43 kWh	>6,000 kWh p.a.

Daily denomination ensures IBT billing is equitably applied for any meter reading period (including NMIs where a customer move-out/move-in occurs), based on an accumulated total of consumption divided by the number of days in the reading period.

The IBT network tariff calculation methodology is as follows:

- a meter read is taken, with total consumption for the number of days within that meter reading period
- for IBT network billing purposes, the energy (kWh) assigned to each block is prorated back to a daily equivalent for that meter reading period
- the bill is then calculated with the component parts being a daily fixed charge, Volume Charge Block 1, Volume Charge Block 2 and Volume Charge Block 3
- daily calculations are then converted back to the total network charge by multiplication by the number of days in the meter reading period
- in relation to TUOS –
 - the TUOS volume component will effectively remain as a flat rate for all IBT tariffs (i.e. same TUOS rate applied in each tariff block or meter reading period)
 - the DLF will be applied to total metered consumption for TUOS.

The IBT bill is calculated as follows:

Component	Calculation
Fixed Charge	Number of days in the period <i>multiplied by</i> fixed charge rate.
Volume Charge Block 1	If equivalent daily consumption is less than the Block 1 daily allowance then: • equivalent daily consumption <i>multiplied by</i> Block 1 rate • multiplied by days in meter reading period • no further calculations required. Or, if equivalent daily consumption exceeds the Block 1 daily allowance then: • Block 1 volume charge calculation applied • proceed to Volume Charge Block 2 calculation.
Volume Charge Block 2	For consumption within the Block 2 allowance then:

Component	Calculation
	equivalent daily consumption <i>less</i> the Block 1 daily allowance <i>multiplied by</i> Block 2 rate (up to the max block 2 threshold) multiplied by days in meter reading period.
Volume Charge Block 3	For any consumption above the aggregate of Block 1 and Block 2 daily allowance then: - remaining consumption above the Block 1 and Block 2 daily allowance <i>multiplied by</i> the Block 3 rate - multiplied by days in meter reading period.

Example:

A residential customer is read quarterly. On this occasion, the customer is read at 90 days with the start read being 123,400 kWh and the end read as 125,200 kWh. This equates to a consumption of 1,800 kWh for that quarter's read.

Currently, the GST exclusive rates for each DUOS component of the IBT Residential East tariff are as follows:

Network Tariff Code	Fixed charge	Block 1 Rate	Block 2 Rate	Block 3 Rate
ERIBT1	\$1.250	\$0.02108	\$0.08486	\$0.12244

Calculation

<i>Equivalent Daily Consumption</i> = consumption <i>divided by</i> the number of days in the read = $1,800/90 = 20.00$. Fixed Charge = $90 \times \\$1.250 = \\112.500 Volume Charge Block 1 = $2.74 \times \\$0.02108 \times 90 = \\5.198 Volume Charge Block 2 = $(16.43 - 2.74) \times \\$0.08486 \times 90 = \\104.556 Volume Charge Block 3 = $(20 - 2.74 - 13.69) \times \\$0.12244 \times 90 = \\$39.334$ Quarterly Bill = \$261.588

The second quarter for this customer shows a total consumption of 200 kWh over 88 days for the period.

Calculation

<i>Equivalent Daily Consumption</i> = consumption <i>divided by</i> the number of days in the read = $200/88 = 2.273$ Fixed Charge = $88 \times \\$1.250 = \\110.000 Volume Charge Block 1 = $2.273 \times \\$0.02108 \times 88 = \\4.216 Volume Charge Block 2 = $0 \times \\$0.08486 \times 88 = \\0 Volume Charge Block 3 = $0 \times \\$0.12244 \times 88 = \\0 Quarterly Bill = \$114.216

Example:

A residential premise (e.g. a summer holiday beach) is read quarterly. On this occasion, the customer has the following consumption pattern.

	Meter reading period (days)	Consumption kWh
Quarter 1	90	1000
Quarter 2	88	0
Quarter 3	93	0
Quarter 4	95	0

The meter reading at the start of quarter 1 is 240,000 kWh and the reading at the end of the fourth quarter is 241,000 kWh.

Calculation – Quarter 1

Equivalent Daily Consumption = consumption divided by the number of days in the read = $1000/90 = 11.11$.
 Fixed Charge = $90 \times \$1.250 = \112.500
 Volume Charge Block 1 = $2.74 \times \$0.02108 \times 90 = \5.198
 Volume Charge Block 2 = $(11.11 - 2.74) \times \$0.08486 \times 90 = \63.925
 Volume Charge Block 3 = $(0) \times \$0.12244 \times 90 = \0
 Quarterly Bill = \$181.623

Calculation – Quarter 2

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/88 = 00.00$.
 Fixed Charge = $88 \times \$1.250 = \110.000
 Volume Charge Block 1 = $(0) \times \$0.02108 \times 88 = \0
 Volume Block 2 = $(0) \times \$0.08486 \times 88 = \0
 Volume Block 3 = $(0) \times \$0.12244 \times 88 = \0
 Quarterly Bill = \$110.000

Calculation – Quarter 3

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/93 = 00.00$.
 Fixed Charge = $93 \times \$1.250 = \116.250
 Volume Charge Block 1 = $(0) \times \$0.02108 \times 93 = \0
 Volume Charge Block 2 = $(0) \times \$0.08486 \times 93 = \0
 Volume Charge Block 3 = $(0) \times \$0.12244 \times 93 = \0
 Quarterly Bill = \$116.250

Calculation – Quarter 4

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/95 = 00.00$.
 Fixed Charge = $95 \times \$1.250 = \118.750
 Volume Charge Block 1 = $(0) \times \$0.02108 \times 95 = \0
 Volume Charge Block 2 = $(0) \times \$0.08486 \times 95 = \0
 Volume Charge Block 3 = $(0) \times \$0.12244 \times 95 = \0
 Quarterly Bill = \$118.750

Annual total = \$526.623.

Appendix 5: CAC connection unit charge examples

Network Tariff Code	Default Distribution Loss Factor		Distribution Use Of System (DUOS)					
			Connection Unit Charge (NDCUC)	Fixed Charge (NDFC)	Capacity Charge (NDKVACC)	Actual Demand Charge (NDKVAADC)	Volume Charge (NDVC)	Excess Reactive Power Charge (NDERPC)
	Value	Code	\$/day per connection unit	\$/day	\$/kVA of AD per month	\$/kVA per month	\$/kWh	\$/excess kVAr per month
EC66T1	Site specific		\$10.389	\$130.366	\$4.478	\$2.507	\$0.00501	\$0.000

Example 1:

Where the customer's:

AD = 3,500 kVA

Connection Units = 11

Actual Demand for the month = 3,000 kVA

Energy for the month = 1,400,000 kWh.

Calculation

Connection Unit Charge = $\$10.389 \times 30 \text{ days} \times 11 \text{ connection units} = \$3,428.370$

Fixed Charge = $\$130.366 \times 30 = \$3,910.980$

Capacity Charge = $\$4.478 \times 3,500 \text{ kVA} = \$15,673.000$

Actual Demand Charge = $\$2.507 \times 3,000 \text{ kVA} = \$7,521.000$

Volume Charge = $\$0.00501 \times 1,400,000 \text{ kWh} = \$7,014.000$

Total monthly DUOS = $\$37,547.35$

Example 2:

Where the customer's:

AD = 4,000 kVA

Connection Units = 0

Actual Demand for the month = 3,900 kVA

Energy for the month = 1,900,000 kWh.

Calculation

Connection Unit Charge = $\$10.389 \times 30 \text{ days} \times 0 \text{ connection units} = \0

Fixed Charge = $\$130.366 \times 30 = \$3,910.980$

Capacity Charge = $\$4.478 \times 4,000 \text{ kVA} = \$17,912.000$

Actual Demand Charge = $\$2.507 \times 3,900 \text{ kVA} = \$9,777.3000$

Volume Charge = $\$0.00501 \times 1,900,000 \text{ kWh} = \$9,519.000$

Total monthly DUOS = $\$41,119.280$

Appendix 6: Seasonal TOU Demand (STOUD) calculation examples

CAC STOUD:

Network Tariff Code	Distribution Use Of System (DUOS)					
	Connection Unit Charge (NDCUC)	Fixed Charge (NDFC)	Capacity Charge Off-peak (NDKVACCOP)	Actual Demand Charge Peak (NDKVADCP)	Excess Reactive Power Charge (NDERPC)	Off-peak (NDVCOP)
	\$/day per connection unit	\$/day	\$/kVA of AD/ month	\$/kVA/month	\$/excess kVAr per month	\$/kWh
EC66TOUT1	\$10.389	\$0.000	\$6.016	\$11.000	\$0.000	\$0.00401

Example:

Where the customer's:

AD = 4,000 kVA

Connection Units = 0

Actual Demand Peak for the month = 3,600 kVA

Actual Demand Off-Peak for the month = 3,900 kVA

Energy for the month = 1,600,000 kWh

Calculation – Month of January (Summer)

Connection Unit Charge = $\$10.389 \times 31 \text{ days} \times 0 \text{ connection units} = \0

Fixed Daily Charge = \$0

Capacity Charge Off-peak = $\$6.016 \times 4,000 \text{ kVA} = \$24,064.000$

Actual Demand Charge Peak = $\$11.000 \times 3,600 \text{ kVA} = \$39,600$

Excess Reactive Power Charge = Does not apply in 2016–17

Volume Charge Off-peak = Does not apply during a summer month

Total monthly DUOS = \$63,664.000

Calculation – Month of September (Non-summer)

Connection Unit Charge = $\$10.389 \times 30 \text{ days} \times 0 \text{ connection units} = \0

Fixed Daily Charge = \$0

Capacity Charge Off-peak = $\$6.016 \times 4,000 \text{ kVA} = \$24,064.000$

Actual Demand Charge Peak = Does not apply during a non-summer month

Excess Reactive Power Charge = Does not apply in 2016–17

Volume Charge Off-peak = $\$0.00401 \times 1,600,000 \text{ kWh} = \$6,416.000$

Total monthly DUOS = \$30,480.000

SAC Large STOUD:

Network Tariff Code	Threshold above which demand charge applies		Distribution Use Of System (DUOS)			
			Fixed Charge (NDFC)	Actual Demand Charge		Volume Charge Off-peak (NDVCOP)
	Peak kW	Off- peak kW		Peak (NDDCP)	Off-peak (NDDCOP)	
			\$/day	\$/kW/ month	\$/kW/ month	\$/kWh
ESTOUDCT1	20	40	\$32.000	\$53.709	\$12.000	\$0.03084

Example 1:

Where the customer's:

Peak demand for the month = 50 kW

Peak energy for the month = 20,000 kWh

Calculation – Month of February (Summer)

Fixed Charge = \$32.000 x 28 days = \$896.000

Actual Demand Charge Peak = \$53.709 x (50 kW – 20 kW) = \$1,611.270

Actual Demand Charge Off-peak = Does not apply during a summer month

Volume Charge Off-peak = Does not apply during a summer month

Total monthly DUOS = \$2,507.270

Example 2:

Where the customer's:

Off-peak demand for the month = 40 kW

Off-peak energy for the month = 25,000 kWh

Calculation – Month of July (Non-summer)

Fixed Charge = \$32.000 x 31 days = \$992.000

Actual Demand Charge Peak = Does not apply during a non-summer month

Actual Demand Charge Off-peak = \$12.000 x (40 kW – 40 kW) = \$0

Volume Charge Off-peak = \$0.03084 x 25,000 kWh = \$771.000

Total monthly DUOS = \$1,763.000

SAC Small STOUD:

Network Tariff Code	Minimum chargeable demand		Distribution Use Of System (DUOS)			
			Fixed Charge (NDFC)	Actual Demand Charge		Volume Charge (NDVC)
	Peak kW	Off- peak kW		Peak (NDDCP)	Off-peak (NDDCOP)	
			\$/day	\$/kW/ month	\$/kW/ month	\$/kWh
ERTOUDCT1	Nil	3	\$0.000	\$72.782	\$13.261	\$0.02000

Example 1:

Where the Residential customer's:

Peak demand for the month = 2 kW

Energy for the month = 500 kWh

Calculation – Month of February (Summer)

Fixed Charge = \$0

Actual Demand Charge Peak = $\$72.782 \times 2 \text{ kW} = \145.564

Actual Demand Charge Off-peak = Does not apply during a summer month

Volume Charge = $\$0.02000 \times 500 \text{ kWh} = \10.000

Total monthly DUOS = \$155.564

Example 2:

Where the Residential customer's:

Peak Demand for the month = 2.725 kW

Energy for the month = 500 kWh

Calculation – Month of July (Non-summer)

Fixed Charge = \$0

Actual Demand Charge Peak = Does not apply during a non-summer month

Actual Demand Charge Off-peak = $\$13.261 \times 3 \text{ kW} = \39.783 (minimum off-peak demand charge applied)

Volume Charge = $\$0.0200 \times 500 \text{ kWh} = \10.000

Total monthly DUOS = \$49.783

Appendix 7: Ergon Energy's Excess kVAr calculation methodology

Permissible kVAr

$$= \text{SQRT}(AD^2 - (AD * \text{Compliant PF})^2)$$

Where AD = Authorised Demand and PF = Power factor

Actual Monthly kVAr

$$= \text{SQRT}(\text{Monthly ATMD kVA}^2 - \text{Correlating kW}^2)$$

Where ATMD = Anytime Maximum Demand

Excess Reactive Power Charge

$$= \text{Max}(0, \text{Actual Monthly kVAr} - \text{Permissible kVAr}) * \text{Excess Reactive Power Charge}$$

Example 1:

Where the customer's:

AD = 6,000 kVA

Compliant PF = 0.95

Monthly ATMD is 5,000 kVA and 4,000 kW

Excess reactive power charge = \$4.000 / excess kVAr

Calculation

Permissible kVAr

$$= \text{sqrt}(6000^2 - (6000 * 0.95)^2)$$

$$= 1,873 \text{ kVAr}$$

Actual Monthly kVAr

$$= \text{sqrt}(5000^2 - 4000^2)$$

$$= 3,000 \text{ kVAr}$$

Excess Reactive Power

$$= \text{Max}(0, 3000 - 1873)$$

$$= 1,127$$

Total monthly Excess Reactive Power Charge

$$= 1,127 \times \$4.000$$

$$= \$4,508$$

Appendix 8: Ergon Energy's kVA calculation methodology

The four metering data quadrants recorded by interval meters are referred to as:

- kW_{Load} the real component of load power consumed by the customer over a given time period (Export kWh - Ei)
- kW_{Gen} the real component of power generated by the customer over a given time period (Import kWh - Bi)
- $kVAr_{Lag}$ (kilovolt-amperes-reactive-hours) is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is lagging the voltage waveform (Export kVArh - Qi)
- $kVAr_{Lead}$ is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is leading the voltage waveform (Import kVArh - Ki).

The kVA algorithm to be used by Ergon Energy is detailed below.

This "Standard" algorithm specifically excludes:

- incidental generation back into the Ergon Energy network
- any Leading VArS generated by customer equipment
- any contribution to lagging VArS by the generator

through modification of interval data as described below,

$$kVA = \sqrt{(kW_{Load})^2 + (kVAr_{Lag})^2} \text{ for each 30 minute interval at a connection point.}$$

For the purposes of network billing, the monthly maximum demand is the maximum kVA of the half hourly values for each interval during the relevant month, calculated as above.

Where Ergon Energy has agreed to the summation of the customer's demand at more than one metering installation for the purposes of determining a diversified demand charge, the billable demand is the vector summation of kWLoad and kVArLag at the individual metering installations. That is, the chargeable demand is calculated for each interval as:

$$kVA = \sqrt{(kW_{Load}^{Meter\ 1} + kW_{Load}^{Meter\ 2} + \dots)^2 + (kVAr_{Lag}^{Meter\ 1} + kVAr_{Lag}^{Meter\ 2} + \dots)^2}$$

Modification of interval data explained

Within a metering interval, there may be both kWLoad and kWGen, where the real demand at a site swings through zero from load to generation or vice-versa. There may also be both kVArLag and kVArLead, where the reactive power swings from lagging to leading power factor, or vice-versa. Depending upon the excitation level of the embedded generator, it will contribute kVAr_{Lag} or kVAr_{Lead}.

With kVA charging for loads, if the load demand were to be directly calculated as

$$kVA = \sqrt{(kW_{Load})^2 + (kVAr_{Lag})^2} \text{ for each 30 minute interval, the kVArLag component may contain a}$$

contribution from the generator. This has the potential to increase the total kVA and kVAr and may create the monthly maximum load demand and a kVAr level that exceeds the permissible quantity.

Clause 6.1.4 of the NER prevents Ergon Energy from charging DUOS for electricity generated into the network. Therefore, the generator's contribution to kVA charges for the load needs to be negated.

For the purposes of Ergon Energy network billing for loads, the interval data is modified so that in any 30 minute interval where $B_i \neq 0$, Q_i is made equal to 0.

With the half hourly interval values of kWLoad and kVArLag modified to remove generator contribution as described the monthly maximum load kVA charge will be based on the vector sum of E_i and Q_i interval data. A load that swings from export to import within an interval would receive a charge based only on the energy exported from the grid to the customer for the interval and would not include any kVArLag contribution by the generator. Where a peak period has been nominated by Ergon Energy the charge would be based on the load based maximum demand occurring during this period.

Appendix 9: TUOS regional indicators

TUOS regional indicators for 2016–17 CAC and SAC		
Transmission Connection Point	TNI	TUOS Region
Alligator Creek	QALC	T2
Louisa Creek	QALH	T2
Alan Sherriff	QASF	T2
Biloela	QBIL	T1
Bowen North	QBNN	T2
Blackwater 132kV (Rolleston)	QBWH	T1
Blackwater 66 kV	QBWL	T1
Gladstone North (Calliope River)	QCAR	T1
Columboola	QCBL	T1
Cardwell	QCDW	T3
Chinchilla	QCHA	T1
Clare South	QCLR	T2
Cairns 132kV	QCNS	T3
Collinsville	QCOL	T2
Cairns 22 kV	QCRN	T3
Dan Gleeson	QDGL	T2
Dysart	QDYS	T1
Egans Hill	QEGN	T1
El Arish	QELA	T3
Edmonton	QEMT	T3
Garbutt	QGAR	T2
Gin Gin	QGNG	T1
Gladstone South	QGST	T1
Innisfail	QINF	T3
Ingham	QING	T3
Kamerunga	QKAM	T3
King Creek	QKCK	T2
Lilyvale 132kV	QLCM	T1
Lilyvale 66 kV	QLIL	T1
Mackay	QMKA	T2
Moura	QMRA	T1
Middle Ridge	QMRG	T1

**TUOS regional indicators for 2016–17
CAC and SAC**

Transmission Connection Point	TNI	TUOS Region
Broadlea	QMRH	T1
Moranbah 11 kV	QMRL	T1
Moranbah 66 kV	QMRN	T1
Mount Isa	Mt Isa	T4
Nebo	QNEB	T2
Newlands	QNLD	T2
Oakey	QOKT	T1
Pandoin	QPAL	T1
Pioneer Valley	QPIV	T2
Proserpine	QPRO	T2
Rockhampton	QROC	T1
Ross	QROS	T2
Stony Creek	QSYC	T2
Teebar Creek	QTBC	T1
Tangkam	QTKM	T1
Tully	QTLL	T3
Tarong 66 kV	QTRL	T1
Turkinje 132kV	QTUH	T3
Turkinje 66 kV	QTUL	T3
Townsville East	QTVE	T2
Townsville South	QTVS	T2
Woolooga	QWLN	T1
Woree	QWRE	T3
Boat Creek	QYAE	T1
Bulli Creek	QBLK	T1
Kemmis	QEMS	T2
Wandoan South	QWSH	T1

Glossary

Abbreviations

ACS	Alternative Control Service
AD	Authorised demand
AER	Australian Energy Regulator
ATMD	Any Time Maximum Demand
CAC	Connection Asset Customer
DH	Demand High Voltage
DL	Demand Large
DLF	Distribution Loss Factor
DM	Demand Medium
DNSP	Distribution Network Service Provider
DS	Demand Small
DUOS	Distribution Use of System
EDNC	Electricity Distribution Network Code
EG	Embedded Generator
Ergon Energy	Ergon Energy Corporation Limited
Excess kVAr	Excess reactive power
FiT	Feed-in tariff
GWh	Gigawatt hour
IBT	Inclining Block Tariff
ICC	Individually Calculated Customer
kV	Kilovolt
kVA	Kilovolt-ampere
kVAr	Kilovolt-ampere reactive
kW	Kilowatt
kWh	Kilowatt hour
LRMC	Long Run Marginal Cost
MWh	Megawatt hour
NEM	National Electricity Market
NER	National Electricity Rules
NMI	National Metering Identifier
p.a.	Per annum
PV	Photovoltaic
SAC	Standard Asset Customer
SBS	Solar Bonus Scheme
SCS	Standard Control Service
STOUD	Seasonal Time-of-Use Demand

STOUE	Seasonal Time-of-Use Energy
TCP	Transmission Connection Point
TNI	Transmission Node Identifier
TNSP	Transmission Network Service Provider
TOU	Time-of-Use
TUOS	Transmission Use of System

Definitions

Actual demand charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures to signal the effect demand has on the shared network and system augmentation. For ICC, CAC and SAC Large the demand used in the calculation of the charge is the maximum demand recorded in any half hour period each month.
Alternative Control Service	A distribution service provided by Ergon Energy that the AER has classified as an Alternative Control Service under the NER. Includes fee based services, quoted services, Default Metering Services and Public Lighting Services.
Any time energy	Is the amount of energy consumed by the customer irrespective of the time of day.
Any Time Maximum Demand (ATMD)	Is the maximum half hourly demand for a customer that occurs at any time within a specified period.
Australian Energy Regulator (AER)	The AER is an independent statutory authority that is part of the Australian Competition and Consumer Commission. The AER is responsible for the economic regulation of electricity networks in the National Electricity Market (NEM). It also monitors the wholesale electricity and gas markets and is responsible for compliance with and enforcement of the National Electricity Law, NER, National Gas Law and Rules, and the National Energy Retail Law and Rules.
Avoided TUOS	The amount paid to an eligible EG for the locational component of prescribed TUOS services that would have been payable by Ergon Energy to a TNSP had the EG not been connected to the distribution network. The methodology Ergon Energy uses to comply with the NER is set out in the <i>Information Guide for Standard Control Services Pricing</i> .
Business customer	Means a customer who is not a residential customer (as defined in the Queensland EDNC).
Capacity charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures. The capacity charge is reflective of the costs associated with the network capacity required by a customer on a long term basis.
Capital contribution	A capital contribution is a prepayment for the provision of direct control services. A capital contribution may be charged to a customer if the new connection or modification for an existing connection is required to the network to accommodate the connection/modification. Ergon Energy's Connection Policy sets out circumstances in which a capital contribution may be required and details how the capital contribution to be charged to a customer is calculated.
Charging parameter	The constituent elements of a tariff (as defined in the NER).

Connection	The physical link to or through a transmission network or distribution network.
Connection Asset Customer (CAC)	Typically reflects those customers: ▪ with required capacity above 1,500 kVA ▪ with energy consumption typically greater than 4 GWh per annum (p.a.) (but less than 40 GWh p.a.), or ▪ with required capacity below 1,500 kVA where: ○ a customer has a dedicated supply system which is quite different and separate from the remainder of the supply network, or ○ inequitable treatment of otherwise comparable customers will arise from the application of the 4 GWh p.a. threshold. The CAC group is further subdivided into categories based on voltage levels as follows: ▪ 66 kV – connected to either a 66 kV substation or a 66 kV line ▪ 33 kV – connected to either a 33 kV substation or a 33 kV line ▪ 22/11 kV Bus – connected to either a 22 kV or 11 kV substation ▪ 22/11 kV Line – connected to either a 22 kV or 11 kV line.
Connection assets	Those components of a transmission or distribution system which are used to provide connection services. Connection assets are those assets required to connect an electrical installation to the shared network and are all the assets from the connection point back up to and including the network coupling point.
Connection point	The agreed point of supply established between the Network Service Provider(s) and another Registered Participant, Non-Registered Customer or franchise customer.
Connection unit charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures to recover the specific value of investment Ergon Energy made in contributing the costs of the connection assets for that customer.
Customer	A person or entity that receives, or wants to receive a supply of electricity for a premises, or any other distribution service from Ergon Energy.
Default Metering Services	A type of Alternative Control Service. Relates to: ▪ Type 5 and 6 meter installation and provision (before 1 July 2015) ▪ Type 5 and 6 meter installation and provision (on or after 1 July 2015), where the replacement meter is initiated by Ergon Energy as a distributor Type 5 and 6 metering maintenance, reading and data services.
Demand	The amount of electricity energy being consumed at a given time measured in either watts (W) or volt amperes (VA). The difference between the two is the power factor.
Distribution Determination	The AER's Distribution Determination sets the revenue and pricing control regime that Ergon Energy must comply with for the current regulatory control period (i.e. 2015–20).
Distribution network	The electrical system used to transport electricity from the high voltage TCP to distribution network users.

Distribution Use of System (DUOS) charge	Component of the network tariffs which covers costs associated with connection services and/or use of the distribution network for the conveyance of electricity (i.e. Standard Control Services).
East Zone	Those areas where the network users are supplied from the distribution system connection to the national grid and have a relatively low distribution cost to supply. The local government areas covered by the East Zone are located in the <i>Information Guide for Standard Control Services</i> .
Embedded Generator (EG)	EGs are those network users that export energy into the distribution system, except for network users with micro-generation facilities of the kind contemplated under Australian Standard (AS) 4777.1 – 2005. EGs are separated into two categories: ▪ EGs that are connected to the distribution system and only generate into the distribution system ▪ EGs that are connected to the distribution system, generate and take load from the system.¹⁴
Energy	The amount of electricity consumed by a consumer (or all customers) over a period of time. Energy is measured in terms of watt hours (Wh), kilowatt hours (kWh), megawatt hours (MWh) or gigawatt hours (GWh).
Excess reactive power charge (Excess kVAr)	A type of charge (charging parameter) included in Ergon Energy's network tariff structures which is applied against the kVAr used by a customer that exceeds what they would be entitled to use at their minimum compliant power factor at authorised demand.
Fee based services	A type of Alternative Control Service which Ergon Energy undertakes at the request of an identifiable customer, retailer or appropriate third party which is levied as a separate charge. The costs of providing the service (and therefore price) can be assessed in advance of the service being requested.
Fixed charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures which is levied on a fixed dollar amount per day or fixed dollar amount per day per device (as is the case for unmetered supply).
Gigawatt hour (GWh)	1,000,000 kilowatt hours.
High Voltage (HV)	Refers to parts of the network that are 11 kV or above.
Inclining Block Tariff (IBT)	A type of network tariff where the price per kWh increases as consumption thresholds are crossed during a particular time period.
Individually Calculated Customer (ICC)	Typically reflects those customers: ▪ with energy consumption typically greater than 40 GWh p.a., or ▪ with energy consumption lower than 40 GWh p.a. where: ○ a customer has a dedicated supply system which is quite different and separate from the remainder of the supply network ○ there are only two or three customers in a supply system, making average prices inappropriate ○ a customer is connected at or close to a TCP, or ○ inequitable treatment of otherwise comparable customers will arise from the application of the 40 GWh p.a. threshold.

¹⁴ The load side will be classified as an ICC, CAC or SAC, and a separate network tariff will apply.

Jurisdictional scheme amount	In respect of a jurisdictional scheme, the amounts a DNSP is required under the jurisdictional scheme obligations to: (a) pay to a person (b) pay into a fund established under an Act of a participating jurisdiction (c) credit against charges payable by a person, or (d) reimburse a person, less any amounts recovered by the DNSP from any person in respect of those amounts other than under the NER (as defined in the NER).
Jurisdictional scheme charges	Component of the network tariff which passes through jurisdictional scheme amounts.
kVA	1,000 Volt-Ampere which is a measure of the apparent power flow which is a measure of the total capacity required to supply a customer's load.
kVAr	1,000 Volt-Ampere reactive which is a measure of reactive power.
kW	1,000 Watts which is a measure of the real component of power being consumed by the consumer's load.
Load factor	Measure of the percentage of time a load is used in any given period. Loads used 24 hours per day, 7 days a week have a load factor of one (1) or 100 per cent.
Low Voltage (LV)	Refers to the sub 11 kV network.
Maximum demand	The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period, such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.
Megawatt hour (MWh)	1,000 kilowatt hours
Mount Isa Zone	Those areas supplied from the isolated Mount Isa system. This zone is not connected to the national grid and would normally be excluded from the application of the NER. However, under the <i>Electricity – National Scheme (Queensland) Act 1997</i> , the Queensland Government has transferred responsibility for the economic regulation of the Mount Isa-Cloncurry supply network to the AER. The local government areas covered by the Mount Isa Zone are located in the <i>Information Guide for Standard Control Services</i> .
National Electricity Market (NEM)	The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.
National Electricity Rules (NER)	Rules made under the National Electricity Law which govern the operation of the NEM.
National Metering Identifier (NMI)	A unique number assigned to each metering installation.
Network capacity	The maximum demand (kVA) that the distribution network can provide for at any one time.
Network tariff	Refers to the price (or tariff) that Ergon Energy sets to recover costs associated with the customer's connection and use of the distribution and transmission network, and jurisdictional scheme amounts. Network tariffs comprise DUOS, TUOS and jurisdictional scheme charges.

Network user	There are four network user groups included in Ergon Energy's network tariff structures – ICCs, CACs, SACs and EGs. For the purposes of our network pricing documents, the term 'network user' refers to both a 'customer' and an 'EG'.
Power factor	The ratio of kW to kVA at a metering point during a defined period.
Public Lighting Services	A type of Alternative Control Service. Relates to the provision, construction and maintenance of public lighting assets owned by Ergon Energy, and emerging public lighting technology. Also encompasses public lighting exit fees.
Public lights – Major	Includes the following lantern types: ▪ Metal Halide – above 125 W ▪ Mercury Vapour – above 125 W ▪ High Pressure Sodium – above 100 W.
Public lights – Minor	Includes the following lantern types: ▪ Compact Fluorescent – all wattages ▪ Fluorescent – all wattages ▪ Metal Halide – up to and including 125 W ▪ Incandescent – all wattages ▪ Low Pressure Sodium – all wattages ▪ LED – all wattages ▪ Mercury Vapour – up to and including 125 W ▪ High Pressure Sodium – up to and including 100 W.
Quoted services	A type of Alternative Control Service. Similar to fee based services, but they are priced on application as the nature and scope of these services is variable and the cost (and therefore price) is specific to the individual requestor's needs.
Residential customer	Means a customer who acquires electricity for domestic use (as defined in the Queensland EDNC).
Revenue cap	The Total Annual Revenue, as determined using the revenue cap formula set out in the Distribution Determination.
SAC Large	Those SACs that typically use between 100 MWh p.a. and 4 GWh p.a.
SAC Small	Those SACs that typically use less than 100 MWh p.a.
Standard Asset Customer (SAC)	Typically reflects those customers with annual energy consumption below 4 GWh p.a. Includes customers with micro-generation facilities (such as small scale photovoltaic generators) of the kind contemplated under AS 4777.1 – 2005. The SAC group is further subdivided into network tariff categories based on whether: ▪ the customer's connection is metered or unmetered ▪ the customer's consumption relates to residential or business use ▪ the customer is taking supply at high voltage or low voltage ▪ the customer's consumption is above or below 100 MWh p.a. ▪ the customer has a meter installed capable of recording demand

	the customer's supply is capable of being controlled by Ergon Energy.
Standard Control Service	A distribution service provided by Ergon Energy that the AER has classified as a Standard Control Service under the NER. Includes network services, some connection services (including small customer connections) and Type 7 metering services. Ergon Energy recovers our costs in providing Standard Control Services through the DUOS component of network tariffs which are billed to retailers.
Summer	The months of December, January and February.
Tariff class	A class of customers for one or more Direct Control Services who are subject to a particular tariff or particular tariffs (as defined in the NER).
Threshold demand	The amount by which a SAC Large customer's metered monthly actual kW maximum demand is adjusted for the purposes of calculating the demand component of their network tariff. The actual demand charge for any time demand tariffs and the peak and off-peak demand charges for the STOUD tariffs are applied to the kW amount by which the recorded monthly maximum demand exceeds the relevant threshold. This demand may occur at any time during the month (actual demand charge and off-peak demand charge) or during a set peak period (peak charge). Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero.
Time-of-Use (TOU)	A type of network tariff where the price per kWh varies according to when the consumption occurs. The TOU tariff may apply a different price during peak and off-peak periods.
Transmission Use of System (TUOS) charge	Component of the network tariff which passes through costs associated with use of the transmission network. This includes designated pricing proposal charges as defined under the NER plus charges levied on Ergon Energy in relation to Chumvale and three Powerlink connection points.
Unmetered	A customer who takes supply where no meter is installed at the connection point.
Volume charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures which is calculated using the customer's metered energy (kWh) consumption. It may be based on a flat rate, an inclining block or TOU charging structure (depending on the customer's applicable network tariff).
West Zone	Those areas outside the East Zone and connected to the national grid, which have a significantly higher distribution cost of supply than the East Zone. The local government areas covered by the West Zone are located in the <i>Information Guide for Standard Control Services</i> .

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